

A DEEP LEARNING-BASED APPROACH FOR CAMERA MOTION CLASSIFICATION

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ABSTRACT

The automatic estimation of the various types of camera motion (*e.g.*, traveling, panning, rolling, zoom...) that are present in videos represents an important challenge for automatic video indexing. Previous research works are mainly based on optical flow estimation and analysis. In this paper, we propose a different, deep learning-based approach that makes it possible to classify the videos according to the type of camera motion. The proposed method is inspired from action recognition approaches and exploits 3D convolutional neural networks with residual blocks. The performances are objectively evaluated on challenging videos, involving blurry frames, fast/slow motion, poorly textured scenes. The accuracy rates obtained (with an average score of 94%) demonstrate the robustness of the proposed model.

Index Terms — Camera motion classification, deep learning, Resnet, 3D CNN

1. INTRODUCTION

The fast growth of available online videos on top of the increase in internet connectivity has reshaped today's media consumption. Video streaming is emerging as the leading media channel and is expected to expand even more in the future. According to Cisco, by 2021, 82% of internet traffic will be dedicated to video. In reaction to such changes, the television broadcasters are adjusting their strategies by converging to automatic and fast video analysis tools to detect valuable sources, prevent fake or manipulated contents, and respect copyright. Within this context, the development of efficient video classification, indexing, and retrieval methodologies becomes a crucial issue. In particular, the camera motion pattern that is present within a given shot represents an important cue, defining a high-level, qualitative description of the shots.

The main categories of camera motion types that are usually present in videos are the following: pan left/right (horizontal camera motion), tilt up/down (rotation in the vertical plane), zoom in/out. Some examples are illustrated in Figure 1.

The camera motion pattern is exploited by TV broadcasters for various purposes. A first application concerns the video content production process itself, which involves different professions, each with its own specificity. Thus, the video shooting is performed by

operators, which are acquiring a large pool of shots, with various types of camera motions.

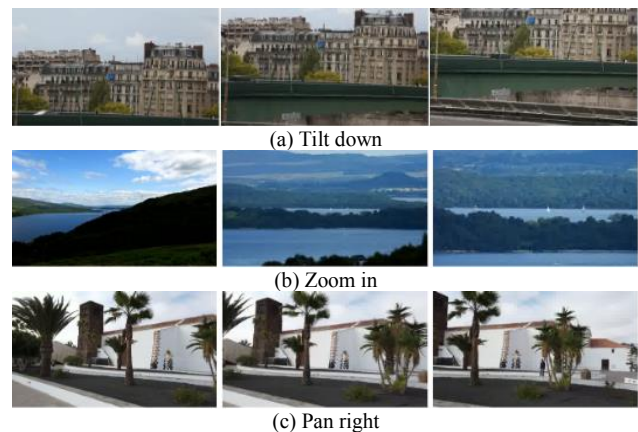


Fig. 1. Examples of the basic camera movement types.

The selection of the most pertinent shot is then done by the video editors, according to esthetic, artistic or operational criteria. Within this context, the camera motion type is an important aspect that needs to be taken into account. However, the related information is lost on the editing side and the video editors have to manually search and select the most convenient shots. Disposing of an automatic motion type classification tool would greatly help them to accelerate the video editing process.

Camera motion classification is also at the core of the archive indexing, browsing and retrieval process. Notably, an important part of the documentalist's job consists of tagging the principal components of a video, in order to classify them properly in the archive, and enable various retrieval and re-use strategies. The camera motion type is part of the mandatory tags that are specified by the documentalists.

In this paper, we explore the use of a 3D CNN model for camera motion classification. Like any deep learning-based approach, the related constraint relies on the availability of an important volume (usually thousands to millions of shots) of labeled samples, which are necessary for setting up a successful training process. However, currently there is no such publicly available data set in the literature, because of the tedious and expensive manual annotation process required to constitute such a corpus.

The fundamental question that we have to solve is then the following: how can we learn a deep CNN model

dedicated to camera motion categorization purposes, with a limited amount of training data?

In order to solve this problem, the first contribution introduced in this paper consists of a transfer learning-based camera motion classification method. The originality of the approach comes from the fact that the initial learning is performed on a data set that is completely different, in terms of target classes, from our camera motion categories. More precisely, we use for training the Kinetics [1] action corpus, which consists of 400 action categories. Such a corpus has nothing to do with our purpose. However, we claim that the derived feature maps capture essential, salient spatio-temporal cues that can be exploited for camera motion classification. A fine-tuning is then applied on a dedicated camera motion data set with a reduced number of items. Our second contribution notably concerns a semi-automatic method that makes it possible to construct a reliable camera motion dataset from general public videos with a minimum amount of human intervention.

Finally, the third contribution concerns the creation of a camera motion evaluation data set that we plan to make available for the research community in order to constitute a reference benchmark corpus. The corpus includes highly challenging videos, acquired in real-life conditions with professional cameras and at various resolutions. It allowed us to assess the robustness and power of generalization of the proposed technique, which yields an average accuracy rate of about 94%. To the very best of our knowledge, our work presents the first data-driven solution to characterize camera motion in videos using a deep 3D convolutional neural network.

The rest of this paper is organized as follows. Section 2 presents a state of the art review, describing camera motion estimation techniques, 3D convolutional neural networks applied to video analysis and available data sets. The proposed method is presented in detail in Section 3. The experimental results, with experimental protocol adopted and objective evaluation are detailed in Section 4. Finally, Section 5 concludes the paper and opens some perspectives of future work.

2. RELATED WORK

2.1. Traditional approaches

Camera motion is an important cue in numerous industrial and research applications and has, therefore, received significant attention in the literature. Traditional approaches are based on the analysis of the motion vectors between successive frames, which serve to determine higher-level parametric motion models, such as affine [2], simplified affine [3], planar perspective or homographies [4] that can globally describe the camera motion. Concerning the motion vectors, they can be determined by simple block matching techniques, dense optical flow approaches or tracking of interest points [5]. The camera motion type is then estimated by comparing the parameters, or corresponding motion vectors, to predefined thresholds. Let us note that the direct estimation of the camera motion is also possible [6].

The availability of a global, parametric motion model creates the premises for determining a more qualitative camera motion model. However, the various heuristics usually employed for their analysis remain too elementary for providing reliable and generic camera motion classification solutions.

2.2. Deep convolutional neural networks techniques

The emergence of deep neural networks has led to a revolutionary break-through, with spectacular performances, in various video analysis applications, including video and object classification, object and action recognition.

The most popular approaches are based on convolutional neural networks (CNNs), which may involve both 2D and 3D (*i.e.*, 2D+t) video representations.

In the case of 2D approaches, it is generally required to include a supplementary neural network, in order to deal with the temporal dimension. Most often, such networks are based on LSTM (*Long Short Time Memory*) [7] or GRU (*Gated Recurrent Unit*) [8] models. On the contrary, 3D CNNs are able to naturally perform a fully spatio-temporal analysis.

Within this context, let us first mention the approach based on 2D CNNs and LSTM [9], which was first explored as a baseline method for visual recognition and description. Simonyan *et al.* [10] introduced the so-called two-stream 2D CNN, which combines RGB frames and optical flow sequences for action recognition in videos. In [11], authors further combined Resnet [12] with two-stream 2D CNNs and proved the effectiveness of this architecture for action recognition.

On the other hand, 3D convolutional networks can intuitively capture salient spatio-temporal features. The principle here consists of the corresponding CNN filters are in this case 3D kernels that jointly act on the spatial and temporal dimensions. The advantage of such an approach comes from the fact the spatio-temporal features are naturally taken into account. On the counterpart, the inclusion of a supplementary dimension leads to an increase in the size of the network parameters, which is usually of an order of magnitude greater than in the 2D case.

The use of 3D CNNs has recently become popular in video classification and outperformed 2D networks. For example, C3D [13] is a VGG-based 3D architecture trained on the Sports-1M dataset. Kay *et al.* [1] introduced the Kinetics dataset and proved its effectiveness to train the C3D network. The Kinetics dataset has 400 action categories and counts for more than 400 samples per category. Similarly, in [14], authors developed an Inception-like 3D topology and achieved state-of-the-art results in action recognition. In [15], [16], [17], the authors conducted several studies related to the 3D Resnet [18] architectures in video recognition. They notably point out that 3D networks require larger learning datasets than 2D CNNs to ensure convergence of the learning process. However, the introduction of large-scale datasets like Kinetics has made it possible to optimize the considerably

high number of parameters involved in 3D CNNs and outperform the related 2D networks performances.

The application of deep neural network approaches to camera motion identification is conditioned by the availability of sufficiently large data sets, required for the learning process.

However, cropping and annotating videos at the frame/shot level is highly time-consuming and requires expert knowledge. As far as we know, the unique open dataset today available is the one released in [19]. It includes the following four categories: static, tilt, pan, and zoom. A number of 2044 videos have been labeled manually, according to the dominant motion in each shot. However, the database presents several limitations. First, the video shots are not trimmed accurately and thus may include some intervals that can skew the classification task. For example, a tilting video of 10s duration can contain 3s of static intervals. Second, only the global label is considered for each type of motion. For example, zoom in and zoom out videos are regrouped within a single category called zoom, while in terms of motion, they exhibit opposite directions. This agnosticism of the categorization with respect to the orientation of motion can raise convergence issues when a learning model is applied. Finally, we have noticed also false labels in some videos, due to human annotation errors. For all these reasons, we have decided not to use this corpus in our work. The proposed solution is described in the following section.

3. PROPOSED METHODOLOGY

3.1. Network architecture

The 3D Resnet network [15] adopted in our work extends to spatio-temporal data the 2D Resnet approach, successfully used in the past for image classification purposes. The underlying principle behind a Resnet network consists of increasing the depth of the network without generating vanishing gradients. To this end, the skip connections reinject the signal into the downstream flow of data. Mathematically, it signifies that instead of learning an original mapping $H(x)$, a Resnet layer approximates the residual mapping $F(x) = H(x) - x$.

We have adopted the extension of Resnet-34 by implementing 3D convolutions. The architecture consists of a first $(7 \times 7 \times 7)$ convolutional layer followed by maxpooling layer and 4 residual units. A global average pooling and a fully connected layer are then considered, succeeded by a softmax activation function (Fig. 2).

We use the basic Resnet block with type ‘A’ shortcut connection [17] and zero padding. The residual blocks are composed of 3, 4, 6 then 3 convolutions respectively. We spatially down-sample the input in the 3rd to 5th bottleneck (stride $s=2$). The size of the input video is $10 \times 120 \times 120 \times 3$.

The considered 3D Resnet architecture is first pre-trained on the Kinetics action dataset, which offers a sufficient quantity and variety of data for achieving a successful training process. However, in order to reach our objectives, a fine-tuning is necessary on a data set with camera motion categories. To this purpose, we introduce a

semi-automatic method, described in the following section that makes it possible to construct such a dedicated camera motion training dataset.

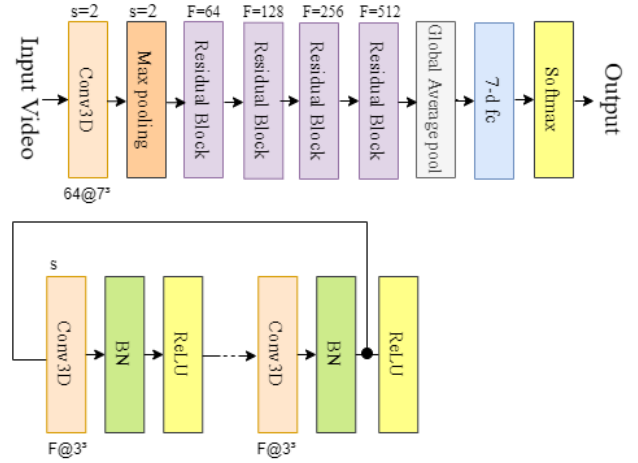


Fig. 2. Illustration of the adopted 3D CNN. The notation $F@H^3$ means F filters of size $H \times H \times H$.

3.2. Camera motion datasets

3.2.1. Camera motion training dataset

Due to the lack of available camera motion datasets, we have constructed our dataset from scratch. We consider the following seven distinct categories: static, pan left (PL), pan right (PR), zoom in (ZI), zoom out (ZO), tilt-up (TU), and tilt down (TD).

First, we have collected the videos randomly from various movies and Youtube [20] videos. We have observed that most shots are static and some of them are panning or tilting. However, zoom is more rarely used. Therefore, we searched for videos including keywords like ‘camera zooming’ which are generally recorded with a professional camera.

We followed the approach introduced in [4] in order to identify the camera motion without any training. We start by extracting interest points using the pyramidal SIFT algorithm [21]. We decided to use SIFT because the method returns a high number of interest points even for low resolution videos. Then, we apply the Lucas-Kanade algorithm [22] in order to determine the interest points displacement vectors.

Based on the motion vectors, the dominant homographic matrix H between two consecutive frames is estimated, using the RANSAC technique [23]. The principle consists in determining the homography which optimally fits, in the mean square error sense, a maximum number of corresponding interest points. The homographic retro-projection error associated to the interest points serves as a criterion to separate between background interest points, which determine the camera motion, and foreground interest points, which belong to foreground objects evolving in the scene that should not be taken into account in the camera motion estimation process.

In order to increase the robustness of the global motion estimation and to detect also non-affine camera motions

such as zoom in/out, we propose to analyze multiple homographies over the same image. Thus, we divide the video frame into four, equal, non-overlapping rectangles. For each rectangular region we compute the homographic matrix in order to determine the inliers/outliers. The interest points assigned to the inliers class satisfy the transformation and belong to the camera/background motion. More precisely, if the homographic retro-projection error, is superior to a threshold (set to 2 pixels in our experiments), we consider the selected keypoints as outliers (*i.e.*, belonging to the foreground) and are not considered for further analysis. Figure 3 illustrates the determined background and foreground points.

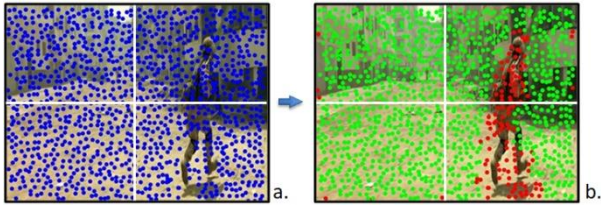


Fig. 3. Automatic separation of background/foreground interest points: (a) Interest point tracking. (b) Interest points assignment to background (blue) / foreground (red) motion

The camera motion is determined by analyzing the inliers interest points' motion vectors. The pan left, pan right, tilt-up, and tilt down are estimated using a majority voting scheme that takes into account the motion vectors magnitude and orientation and compares them with pre-defined values, characteristic for each camera motion types. In order to validate a specific camera motion the results of the estimation should be consistent in all four image regions. For example, for zoom-out the motion vectors from each image region should point to its associated video frame corner, while for zoom-in the vectors are directed toward the frame center.

The considered method presents some limitations. For example, in high-speed videos or outdoor shots with a lack of texture, the algorithm fails to detect keypoints. We encounter additional problem in close-up shots, where the foreground object occupy an important area of the scene, and blurry shots, where is difficult to reliably detect the interest points. In such cases, the video is classified as unknown and is not included in the dataset. Besides, the Lukas-Kanade tracking system cannot make the difference between static and slow-motion video segments. As a consequence, the process required some human intervention (for less than 5% of the videos) to delete the outliers in the static class. However, let us underline that the goal of this method is solely related to the construction of a reliable training set that can serve to learn the 3D Resnet network described in Section 3.1, and which includes only correctly categorized video segments. To this purpose, the method was successful and we managed to acquire a total number of 700 videos, with 100 videos of about 2s duration in each category.

3.2.2. Validation dataset

To fairly evaluate our method we have considered a validation dataset that we have acquired ourselves under different conditions. It can include relative low quality videos (blur/camera shake), illumination variations or shadows. More importantly, we have included a great diversity in the scenes by varying the shot type (indoor/outdoor, close up/wide shots), the meteorological conditions (sun/rain) or time (day/night). The recordings have been performed with the help of professional cameras (Canon 5D Mark II and Sony Z280) with tripods. Several shots have been acquired with smartphones in order to include some hand-shaken videos. We have also considered various resolutions (from 1280 x 720 to 3840 x 2160) and frame rates (25 to 50 fps). Each video lasts around 5s and can contain several camera movements, separated by static parts. Table 2 summarizes the resulting numbers of video segments (*i.e.*, video intervals described by a unique camera motion) per category obtained.

Table 2. The number of videos per category in the validation dataset

Category	S	PL	PR	TU	TD	ZI	ZO
No. of video segments	450	369	422	261	263	251	252

Figure 4 illustrates some example of video sequences from the validation dataset.

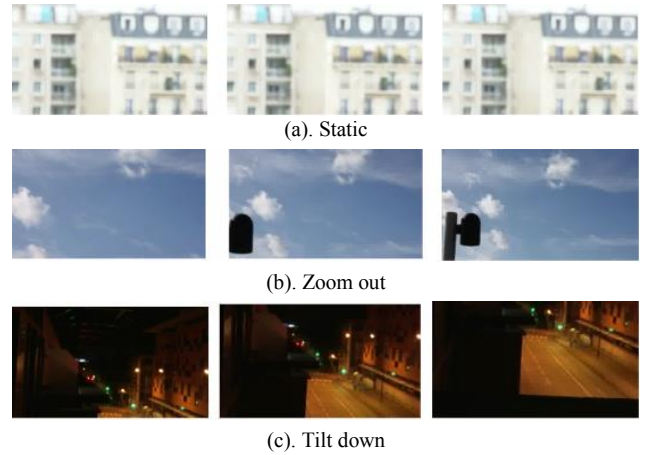


Fig. 4. Examples from the validation dataset.

4. EXPERIMENTAL RESULTS

We first train the model on action recognition using the Kinetics dataset [12]. The backpropagation algorithm minimizes the cross-entropy error, considered as loss function. For optimization, we have adopted a stochastic gradient descent with momentum. The weight decay was set to 0.001 and the momentum to 0.9. The learning rate started from 0.9 and was divided by 10 when the test loss saturated.

Then, we retrain the model starting from the 3rd residual block with the camera motion dataset. For fine-tuning, we set the learning rate to 0.001 and the weight decay to 1e-5. The learning process has been carried out on

an Nvidia GTX 1080 GPU with batch size of 32. We randomly crop each video spatially and temporally to fit the network's input size $10 * 120 * 120 * 3$.

To increase scalability, we have also applied a data augmentation technique. To this purpose, we have reversed the order of the frames in each video and generate a new video sample with the opposite label (*e.g.*, if we reverse the frames of a zoom-in video we generate a new zoom-out video). This technique makes it possible not only to double the size of the dataset but also increases the generality of the model.

Figure 5 shows the loss and accuracy curves obtained for four different conditions.

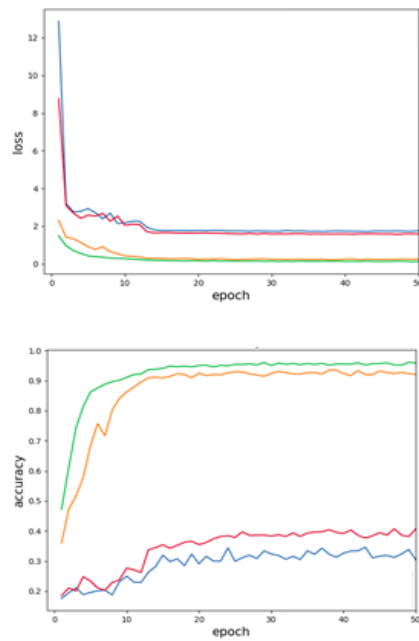


Fig. 5. Comparison of the different configurations: (a) The loss variation (b) The accuracy variation. (Blue: Resnet trained from scratch, Red: Resnet + reverse frames, Orange: Resnet + finetuning, Green: Resnet + finetuning + reverse frames)

The baseline results are obtained by training Resnet-34 from scratch, without any preliminary training, with our training dataset described in Section 3.2.1. Without surprise, this approach resulted in poor results due to the low number of training samples. Augmenting the data by reversing the frames slightly improves the results but does not exceed an accuracy rate of 40%. These results confirm our intuition that such limited datasets cannot ensure a successful learning process. Let us now analyze the performances obtained for the transfer learning approach, where the 3D Resnet is first pre-trained on the Kinetics dataset and then fine-tuned on the augmented training set.

In the first time, we have examined the results obtained on the training data set. For this purpose, we have split the data into training and test sub-sets with a training/testing ratio of 80/20% and 5 cross-validation steps. In this case, the approach reaches 97% of accuracy.

Concerning the validation data set, since a given video may contain multiple camera motions, we have jointly performed segmentation and labeling. For this purpose, we have applied a sliding window technique with overlapping

clips and stride 1, which means that the sliding window is successively slid by one frame. The size of the window is of intervals of 10 successive frames. At each frame position, the trained network estimates the class probabilities. As the video may contain several camera movements we use the static segments (a segment is a set of 2 clips or higher) to identify the end of a movement section. Then, we average the scores and the decision is taken based on a majority voting scheme over all frames of a movement section. The class with the highest probability is estimated as the correct camera motion.

The resulting global, average accuracy rate obtained is 94%. These results, obtained on such challenging videos, fully demonstrate the pertinence of the proposed approach.

In a finer analysis, Figure 6 presents the confusion matrix obtained on the validation dataset.

Static	0.97	0	0.01	0.01	0	0	0.01
PL	0.03	0.91	0.01	0	0.01	0.02	0.02
PR	0	0.02	0.94	0.02	0.01	0	0.01
TU	0.01	0	0	0.98	0	0.01	0
TD	0	0	0	0	0.98	0.02	0
ZI	0.02	0.03	0.01	0.01	0	0.93	0
ZO	0.01	0.02	0.01	0.01	0.02	0.02	0.9
	Static	PL	PR	TU	TD	ZI	ZO

Fig. 6. Confusion matrix of the validation dataset

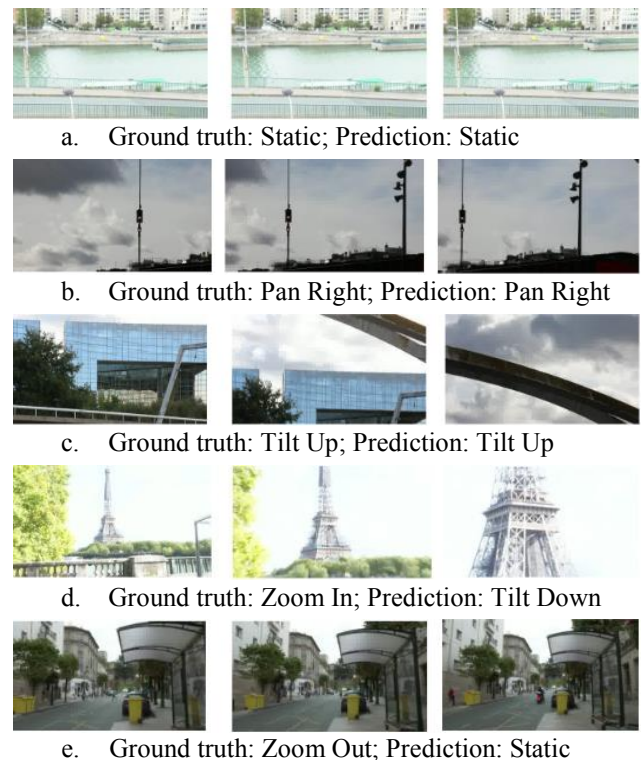


Fig. 7. Examples of recognition results on the validation dataset.

We can observe that some errors are occurring when the zooming is either very slow or very fast. One explanation of this behavior may be the lack of diversity in the zoom class in the train dataset (Fig.7).

5. CONCLUSION AND PERSPECTIVES

In this paper, we have introduced a data-driven camera motion classification method. The approach is based on a deep 3D Resnet model, under a transfer learning paradigm. In order to overcome the difficulties related to the poor availability of training/test camera motion data sets, we have proposed a transfer learning approach, which consists first in pre-training the model on a different purpose data set, related to action recognition, which is afterward finetuned on a lower scale, dedicated corpus. We have also introduced a semi-automatic technique for constructing such a corpus, starting from general-purpose videos. Finally, in order to test and validate our approach, we have acquired a validation dataset, acquired in real-life conditions with professional cameras and involving highly challenging videos that we intend to make available to the research community. The experimental results obtained fully validate the proposed method, with an average accuracy rate of 94%. Our perspectives of future work mainly concern the extension of our approach with the help of active learning techniques that can further speed-up the training process and enhance the related performances.

ACKNOWLEDGMENT

This work has been carried out within the framework of the AITV (*Artificial Intelligence for Television*) established between Télécom SudParis and France Télévisions.

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